AJ_24038155_dialect_v1.5

Full integrated Cambridge-Style Pseudocode specification

Integrated specification: This document incorporates the complete v1.0 dialect and the approved v1.1, v1.2, v1.3, v1.4 and v1.5 language additions into a single normative specification.

1. Language Overview

The AJ_24038155 Dialect is a structured pseudocode language intended for:

- representing algorithms generated from natural-language tasks;
- deterministic parsing into an Abstract Syntax Tree;
- conversion into a Control Flow Graph;
- visual generation of flow diagrams;
- deterministic translation into Python;
- safe behavioural evaluation. The language uses explicit block terminators and does not rely on indentation for program structure.

Keywords are case-insensitive during parsing but are presented in uppercase as the canonical format.

Identifiers remain case-sensitive.

2. Lexical Rules

2.1 Identifiers

An identifier must begin with a letter or underscore.

```
identifier ::= letter , { letter | digit | "_" } ;
```

Examples:

total

item_count

_grid

value2

Invalid examples:

2value

item-count

total value

2.2 Integer Literals

```
integer_literal ::= [ "-" ] , digit , { digit } ;
```

Examples:

0

12

-5

1000

2.3 Float Literals

```
float_literal ::= [ "-" ] , digit , { digit } , "." , digit , { digit } ;
```

Examples:

0.5

3.14

-10.25

2.4 String Literals

Strings use the STRING constructor with quoted contents.

```
string_literal ::= "STRING" , "(" , quoted_string , ")" ;
```

Examples:

```
STRING("Hello")
STRING("Enter a number")
STRING("The result is {result}")
```

Escape sequences may be used:

```
\"
```

```
\\
```

```
\n
```

```
\t
```

Example:

```
STRING("He said \"Hello\"")
```

2.5 Character Literals

Characters use the CHAR constructor and must contain exactly one character or one valid escape sequence.

```
char_literal ::= "CHAR" , "(" , quoted_character , ")" ;
```

Examples:

```
CHAR("a")
CHAR("7")
CHAR("\n")
```

2.6 Boolean Literals

```
boolean_literal ::= "BOOLEAN" , "(" , ( "TRUE" | "FALSE" ) , ")" ;
```

Canonical examples:

```
BOOLEAN(TRUE)
BOOLEAN(FALSE)
```

2.7 Native Null Literal

NONE is the native null literal in v1.1.

```
none_literal ::= "NONE" ;
```

Keywords are case-insensitive during parsing, therefore NONE, None and none denote the same literal.

Canonical output uses uppercase NONE.

Python translation:

```
NONE
```

translates deterministically to:

None

NONE is a value. It is not an identifier and is not a function call.

The historical compatibility form CALL NULL() remains accepted for existing v1.0 material. New canonical v1.1 pseudocode should use NONE.

2.8 Comments and Import Directives

Single-line comments begin with COMMENT. AJ v1.5 distinguishes ordinary non-executable comments from a narrow family of exact executable import directives.

```
comment ::=
    ordinary_comment
```

```
  | import_directive ;

ordinary_comment ::=
    "COMMENT" , comment_text , newline ;

import_directive ::=
    "COMMENT" , "import" , module_reference , [ "AS" , identifier ] , newline
  | "COMMENT" , "from" , module_reference , "import" , import_target_list , newline ;

module_reference ::=
    identifier , { "." , identifier } ;

import_target_list ::=
    "*"
  | identifier , { "," , identifier } ;
```

Ordinary example:

```
COMMENT Calculate the running total
```

Executable import examples:

```
COMMENT import string
COMMENT import numpy AS np
COMMENT from math import sqrt
COMMENT from math import sqrt, floor
COMMENT from math import *
```

Only a complete COMMENT payload matching one of these import forms is executable. Every other COMMENT remains non-executable metadata. The wildcard form is canonically `from <module> import *`; `from * import <module>` is not valid AJ v1.5 syntax.

Recognised import directives are represented by `ImportDirectiveNode`; ordinary comments remain `CommentNode`.

Python lowering is direct and lexical:

```
COMMENT import string
-> import string

COMMENT import numpy AS np
-> import numpy as np

COMMENT from math import sqrt, floor
-> from math import sqrt, floor

COMMENT from math import *
-> from math import *
```

No import may be inferred from arbitrary prose or from an unresolved identifier.

2.8.1 Standard-Library Punctuation Adapter

AJ v1.5 defines the exact case-sensitive identifier:

```
string_method_punctuation
```

as the punctuation-character collection provided by Python's standard `string` module.

Deterministic Python lowering:

```
string_method_punctuation
-> string.punctuation
```

The module dependency must still be declared explicitly where required, for example:

```
COMMENT import string
```

This rule does not assign semantics to other punctuation-related prose, predicates or identifiers.

2.9 Whitespace and Newlines

Spaces and indentation are not semantically significant except inside string literals.

Each statement must end with a newline.

Block structure is determined by explicit keywords such as:

```
ENDIF
ENDWHILE
ENDFOR
ENDFUNCTION
```

3. Operator Rules

3.1 Assignment

Assignment uses <-.

```
variable <- value
```

Example:

```
total <- 0
name <- STRING("Alan")
```

3.2 Arithmetic Operators

```
+

-

*

/

INTEGER DIVISION
MODULUS
EXP
```

Examples:

```
sum <- a + b
difference <- a - b

product <- a * b
quotient <- a / b
whole <- a INTEGER DIVISION b
remainder <- a MODULUS b
power <- x EXP y
```

3.3 Comparison Operators

```
==

!=

<

>

<=

>=
```

Examples:

x == y

x != y

x= y

3.4 Boolean Operators

```
NOT

AND
OR
XOR
```

Examples:

x > 0 AND x < 10

NOT finished

left OR right

flag1 XOR flag2

3.5 Operator Precedence

Operators are evaluated in the following order, from highest to lowest:

1. **Parentheses**
2. **Function and collection access**
3. **EXP**
4. **Unary minus**
5. ***, /, INTEGER DIVISION, MODULUS**
6. **+, -**
7. **==, !=, <, >, <=, >=**
8. **NOT**
9. **AND**
10. **XOR**
11. **OR**

Parentheses should be used where ambiguity may occur.

4. Data Types and Literals

4.1 Supported Primitive Types

```
INTEGER
FLOAT
STRING
CHAR
BOOLEAN
```

4.2 Integer Literal

```
INTEGER(5)
```

```
INTEGER(-20)
```

Formal rule:

```
typed_integer ::= "INTEGER" , "(" , integer_literal , ")" ;
```

4.3 Float Literal

```
FLOAT(3.14)
FLOAT(-0.5)
```

Formal rule:

```
typed_float ::= "FLOAT" , "(" , float_literal , ")" ;
```

4.4 Tuple Literal

```
TUPLE(a, b)
TUPLE(INTEGER(1), STRING("A"))
```

Formal rule:

```
tuple_literal ::=
```

“TUPLE” ,

“(” ,

[expression , { “,” , expression }] ,

“)” ;

4.5 Homogeneous Typed Lists

A homogeneous list contains values of one declared type.

Examples:

```
LIST OF INTEGER [1, 2, 3]
LIST OF STRING [STRING("A"), STRING("B")]
LIST OF BOOLEAN [BOOLEAN(TRUE), BOOLEAN(FALSE)]
```

Formal rule:

```
typed_list_literal ::=
```

“LIST OF” ,

type_name ,

“[” ,

[expression , { “,” , expression }] ,

“]” ;

All values must be compatible with the declared type.

4.6 Mixed Lists

Mixed lists use the LIST keyword.

Example:

```
LIST [INTEGER(5), STRING("hello"), CHAR("a"), BOOLEAN(TRUE)]
```

Formal rule:

```
mixed_list_literal ::=
```

“LIST” ,

“[” ,

[expression , { “,” , expression }] ,

“]” ;

4.7 Empty Values

Supported empty values are:

```
EMPTY LIST
EMPTY DICTIONARY
EMPTY STRING
EMPTY SET
EMPTY TUPLE
```

Formal rule:

```
empty_literal ::=
```

“EMPTY LIST”

```
| "EMPTY DICTIONARY"
| "EMPTY STRING"
| "EMPTY SET"
| "EMPTY TUPLE" ;
```

4.8 Native Null Literal

NONE represents the absence of a value.

Canonical syntax:

```
NONE
```

Formal rule:

```
none_literal ::= "NONE" ;
```

Python translation:

NONE -> None

5. Program Structure

5.1 Program

A program contains one or more declarations or statements.

```
program ::= { declaration | statement | comment } ;
```

A program should normally contain at least one function when it is intended for automated testing.

5.2 Function Definition

Canonical syntax:

```
FUNCTION name ACCEPTS(parameter1, parameter2)
```

statements

```
ENDFUNCTION
```

Example:

```
FUNCTION calculate_total ACCEPTS(numbers)
total <- 0

FOR EACH number IN numbers DO
total <- total + number
ENDFOR

RETURN total
ENDFUNCTION
```

Formal rule:

```
function_definition ::=
"FUNCTION" ,
identifier ,
"ACCEPTS" ,
"(" ,
[ parameter_list ] ,
")" ,
newline ,
function_body ,
"ENDFUNCTION" ,
newline ;

function_body ::= { function_body_item } ;

function_body_item ::=
statement
| function_definition
| comment ;
```

5.2.1 Nested/local function declarations A FUNCTION … ENDFUNCTION declaration may appear inside the body of another function. A function declared in this way is a nested function, also called a local function.

Nested functions use the existing FUNCTION syntax and may themselves contain nested function declarations. The nested function name belongs to the lexical scope of the enclosing function and must not be promoted to module scope.

```
FUNCTION outer ACCEPTS(value)
  FUNCTION inner ACCEPTS(item)
    RETURN item
  ENDFUNCTION

  RETURN CALL inner(value)
ENDFUNCTION
```

A nested function may call itself recursively by name and may read values visible in an enclosing lexical scope. Assignment to a simple identifier inside the nested function follows the nested function local-binding rule; v1.2 introduces no NONLOCAL-style rebinding construct.

v1.2 adds declaration nesting only. It does not introduce callbacks, first-class function values, anonymous functions, or function- reference expressions.

Python translation preserves lexical nesting. A nested AJ function compiles to a nested Python def at the corresponding indentation; the compiler must not hoist it merely to simplify code generation.

5.3 Parameter List

```
parameter_list ::=
```

parameter ,

{ “,” , parameter } ;

A parameter may optionally declare a type.

Examples:

```
FUNCTION process ACCEPTS(value)
FUNCTION process ACCEPTS(value AS INTEGER)
```

Formal rule:

```
parameter ::=
identifier ,
```

[“AS” , type_name] ;

5.4 Return Statement

```
RETURN expression
```

or:

```
RETURN
```

Formal rule:

```
return_statement ::=
```

“RETURN” ,

[expression] ,

```
newline ;
```

A value-less return is permitted.

5.5 Procedure and Function Calls

A call may be used as an expression:

```
result <- CALL calculate_total(numbers)
```

or as a standalone statement:

```
CALL display_result(result)
```

Formal rules:

```
call_expression ::=
```

“CALL” ,

```
identifier ,
```

“(” ,

```
[ argument_list ] ,
```

“)” ;

```
call_statement ::=
```

call_expression ,

```
newline ;
```

```
argument_list ::=
expression ,
```

{ “,” , expression } ;

5.6 Explicit Runtime Casts

Runtime conversion between primitive values uses CALL followed by a primitive cast target.

Canonical syntax:

```
CALL INTEGER(expression)
CALL FLOAT(expression)
CALL STRING(expression)
```

The preceding CALL distinguishes a runtime cast from the existing typed-literal constructors.

Examples:

```
INTEGER(23)
CALL INTEGER(value)
FLOAT(3.5)
CALL FLOAT(value)
STRING("23")
CALL STRING(value)
```

Formal rule:

```
cast_expression ::=
    "CALL" , cast_type , "(" , expression , ")" ;

cast_type ::=
    "INTEGER"
    | "FLOAT"
    | "STRING" ;
```

Each cast accepts exactly one argument. Invalid INTEGER or FLOAT conversions produce a runtime error and may be handled by TRY-CATCH. CALL STRING(expression) produces the deterministic textual representation of the evaluated value.

Python translation:

```
CALL INTEGER(value) -> int(value)
CALL FLOAT(value) -> float(value)
CALL STRING(value) -> str(value)
```

The compiler must not insert casts implicitly. A cast occurs only where the source explicitly contains the corresponding CALL expression.

5.7 Program Termination with QUIT

Canonical syntax:

```
QUIT
```

QUIT immediately terminates execution of the current program. It is distinct from RETURN, BREAK and CONTINUE. RETURN exits the current function and remains invalid at top level; BREAK exits the nearest loop; CONTINUE starts the next loop iteration.

Formal rule:

```
quit_statement ::= "QUIT" , newline ;
```

Python translation:

```
QUIT -> quit()
```

A QUIT statement is terminal in the control-flow graph and must not fall through to a following executable statement.

5.8 Runtime Type Inspection with ISINSTANCE

AJ v1.4 adds explicit runtime type inspection corresponding to Python `isinstance(...)`.

Canonical syntax:

```
CALL ISINSTANCE(value, STRING)
CALL ISINSTANCE(value, (INTEGER, FLOAT))
```

The result is BOOLEAN. The type-reference operand is syntax and does not make AJ type names ordinary first-class values.

Formal rules:

```
isinstance_expression ::=
    "CALL" , "ISINSTANCE" , "(" , expression , "," , type_reference , ")" ;

type_reference ::=
    runtime_type
    | "(" , runtime_type , { "," , runtime_type } , ")" ;

runtime_type ::=
    "INTEGER"
    | "FLOAT"
    | "STRING"
    | "BOOLEAN"
    | "LIST"
    | "DICTIONARY"
    | "SET"
    | "TUPLE"
    | identifier ;
```

Built-in Python translation:

```
INTEGER -> int
FLOAT -> float
STRING -> str
BOOLEAN -> bool
LIST -> list
DICTIONARY -> dict
SET -> set
TUPLE -> tuple
```

Examples:

```
CALL ISINSTANCE(value, INTEGER) -> isinstance(value, int)
CALL ISINSTANCE(value, STRING) -> isinstance(value, str)
CALL ISINSTANCE(value, (INTEGER, FLOAT)) -> isinstance(value, (int, float))
```

Runtime type inspection is explicit. The compiler must not insert an ISINSTANCE test that is absent from the source.

6. Input and Output

6.1 Input

Input must contain a prompt expression.

```
age <- INPUT(STRING("Enter your age"))
```

Formal rule:

```
input_expression ::=
```

“INPUT” ,

“(” ,

```
expression ,
```

“)” ;

6.2 Output

```
OUTPUT expression
```

Examples:

```
OUTPUT result
OUTPUT STRING("The value is {value}")
```

Formal rule:

```
output_statement ::=
```

“OUTPUT” ,

```
expression ,
newline ;
```

String placeholders use braces:

```
STRING("The result is {result}")
```

The referenced identifier must exist in the current scope.

7. Assignment Rules

7.1 Standard Assignment

```
identifier <- expression
```

Example:

```
count <- 0
```

7.2 Collection Element Assignment

```
list[index] <- value
dictionary[key] <- value
```

Examples:

```
numbers[0] <- 10
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

Formal rule:

```
assignment_statement ::=
```

assignable ,

“<-” ,

```
expression ,
newline ;
```

```
assignable ::=
```

identifier

```
| indexed_access ;
```

8. Expressions

8.1 Expression Grammar

```
expression ::= or_expression ;
```

```
or_expression ::=
```

xor_expression ,

{ “OR” , xor_expression } ;

```
xor_expression ::=
```

and_expression ,

{ “XOR” , and_expression } ;

```
and_expression ::=
```

not_expression ,

{ “AND” , not_expression } ;

```
not_expression ::=
```

“NOT” , not_expression

```
| comparison_expression ;
```

```
comparison_expression ::=
```

additive_expression ,

[comparison_operator , additive_expression] ;

```
additive_expression ::=
```

multiplicative_expression ,

{ (“+” | “-”) , multiplicative_expression } ;

```
multiplicative_expression ::=
```

power_expression ,

{

```
(
```

“*”

```
| "/"
| "INTEGER DIVISION"
| "MODULUS"
```

),

power_expression

};

```
power_expression ::=
```

unary_expression ,

[“EXP” , power_expression] ;

```
unary_expression ::=
```

“-” , unary_expression

```
| primary_expression ;
primary_expression ::=
```

literal

```
| identifier
| indexed_access
| member_call_expression
| call_expression
| cast_expression
| isinstance_expression
| input_expression
| length_expression
| join_expression
| dictionary_get_expression
| random_expression
| "(" , expression , ")" ;
```

8.2 Comparison Operator

```
comparison_operator ::=
```

“==”

```
| "!="
| "<"
| ">"
| "<="
| ">=" ;
```

9. Conditional Rules

9.1 IF Statement

```
IF condition THEN
```

statements

```
ENDIF
```

Formal rule:

```
if_statement ::=
```

“IF” ,

```
expression ,
```

“THEN” ,

newline ,

{ statement | comment } ,

{ else_if_clause } ,

[else_clause] ,

“ENDIF” ,

`newline ;`

9.2 ELSE IF Clause

`ELSE IF condition THEN`

statements

Formal rule:

`else_if_clause ::=`

“ELSE IF” ,

`expression ,`

“THEN” ,

newline ,

{ statement | comment } ;

9.3 ELSE Clause

`ELSE`

statements

Formal rule:

`else_clause ::=`

“ELSE” ,

newline ,

{ statement | comment } ;

9.4 Full Example

```
IF score >= 70 THEN
grade <- STRING("Distinction")
ELSE IF score >= 60 THEN
grade <- STRING("Merit")
```

```
ELSE
grade <- STRING("Pass")
ENDIF
```

10. Loop Rules

10.1 WHILE Loop

`WHILE condition DO`

statements

`ENDWHILE`

Formal rule:

`while_statement ::=`

“WHILE” ,

`expression ,`

“DO” ,

newline ,

{ statement | comment } ,

“ENDWHILE” ,

`newline ;`

10.2 Range Loop

Range loops are end-exclusive.

Therefore:

`FOR i <- IN RANGE 0 TO n DO`

iterates through:

0, 1, 2, ..., n - 1

Canonical syntax:

`FOR identifier <- IN RANGE start TO end DO`

statements

`ENDFOR`

Optional step:

`FOR i <- IN RANGE 10 TO 0 STEP -1 DO`

statements

```
ENDFOR
```

Formal rule:

```
range_for_statement ::=
```

“FOR” ,

```
identifier ,
```

“<-” ,

“IN RANGE” ,

```
expression ,
```

“TO” ,

```
expression ,
```

[“STEP” , expression] ,

“DO” ,

newline ,

{ statement | comment } ,

“ENDFOR” ,

```
newline ;
```

A step value must not equal zero.

10.3 FOR EACH Loop

```
FOR EACH item IN collection DO
```

statements

```
ENDFOR
```

Formal rule:

```
for_each_statement ::=
```

“FOR EACH” ,

```
identifier ,
```

“IN” ,

```
expression ,
```

“DO” ,

newline ,

{ statement | comment } ,

“ENDFOR” ,

```
newline ;
```

10.4 BREAK

```
BREAK
```

Formal rule:

```
break_statement ::=
```

“BREAK” ,

```
newline ;
```

```
BREAK is valid only inside:
```

```
WHILE
FOR
FOR EACH
```

10.5 CONTINUE

The dialect includes an optional CONTINUE statement.

```
CONTINUE
```

Formal rule:

```
continue_statement ::=
```

“CONTINUE” ,

```
newline ;
```

```
CONTINUE is valid only inside a loop.
```

11. List and Collection Rules

11.1 Indexing

collection[index]

Examples:

numbers[0]

grid[row][column]

Formal rule:

```
indexed_access ::=
```

primary_identifier ,

{ “[” , expression , ”]” } ;

```
primary_identifier ::= identifier ;
```

Indexes begin at zero.

11.2 Length

LENGTH OF collection

Formal rule:

```
length_expression ::=
```

“LENGTH OF” ,

expression ;

11.3 Append

```
APPEND value TO collection
```

Formal rule:

```
append_statement ::=
```

“APPEND” ,

```
expression ,
```

“TO” ,

```
expression ,
newline ;
```

11.4 Insert

```
INSERT value INTO collection AT index
```

Formal rule:

```
insert_statement ::=
```

“INSERT” ,

```
expression ,
```

“INTO” ,

```
expression ,
```

“AT” ,

```
expression ,
newline ;
```

11.5 Remove First Item

```
REMOVE FIRST ITEM FROM collection
```

Formal rule:

```
remove_first_statement ::=
```

"REMOVE FIRST ITEM FROM" ,

```
expression ,
newline ;
```

11.6 Remove Last Item

```
REMOVE LAST ITEM FROM collection
```

Formal rule:

```
remove_last_statement ::=
```

"REMOVE LAST ITEM FROM" ,

```
expression ,
newline ;
```

11.7 Remove Specific Item

```
REMOVE value FROM collection
```

Formal rule:

```
remove_value_statement ::=
```

"REMOVE" ,

```
expression ,
```

"FROM" ,

```
expression ,
newline ;
```

11.8 Join

```
JOIN(collection, separator)
```

Example:

```
result <- JOIN(characters, EMPTY STRING)
```

Formal rule:

```
join_expression ::=
```

“JOIN” ,

“(” ,

```
expression ,
```

“,” ,

```
expression ,
```

“)” ;

11.9 Two-Dimensional List Creation

Canonical form:

```
grid <- EMPTY LIST
r <- 0

WHILE r < rows DO
row <- EMPTY LIST
c <- 0

WHILE c < cols DO
APPEND fill_value TO row
c <- c + 1
ENDWHILE

APPEND row TO grid
r <- r + 1
ENDWHILE
```

No special two-dimensional list syntax is required. Nested lists are constructed using ordinary list

operations.

12. Set Rules

12.1 Set Literal

```
SET [value1, value2, value3]
```

Formal rule:

```
set_literal ::=
```

“SET” ,

“[” ,

[expression , { “,” , expression }] ,

“]” ;

12.2 Set Operations

```
UNION(set1, set2)
INTERSECTION(set1, set2)
DIFFERENCE(set1, set2)
```

Formal rules:

```
set_operation ::=
```

“UNION” , “(” , expression , “,” , expression , “)”

```
| "INTERSECTION" , "(" , expression , "," , expression , ")"
| "DIFFERENCE" , "(" , expression , "," , expression , ")" ;
```

12.3 Membership

value IN collection

value NOT IN collection

These are treated as comparison expressions.

13. Dictionary Rules

13.1 Dictionary Creation

```
visited <- EMPTY DICTIONARY
```

13.2 Dictionary Assignment

```
dictionary[key] <- value
```

Example:

```
visited[TUPLE(nr, nc)] <- TUPLE(r, c)
```

13.3 Dictionary Retrieval

Canonical v1.1 syntax:

```
value <- dictionary.GET(key)
```

Example:

```
node <- visited.GET(node)
```

Dictionary retrieval uses the general member-method invocation notation.

13.4 Dictionary Retrieval With Default

Optional syntax:

```
value <- dictionary.GET(key, default_value)
```

When resolved as the dictionary intrinsic, GET accepts exactly one or two arguments:

```
dictionary.GET(key)
dictionary.GET(key, default_value)
```

Without a default value, a missing key produces a runtime error.

13.4.1 Semantic Resolution GET is not globally interpreted as dictionary retrieval by spelling alone.

```
The source form receiver.GET(...) is first parsed as an ordinary member-method invocation. I
resolves to the dictionary intrinsic only when static evidence establishes that receiver is
dictionary-backed, for example:
```

```
frequency <- EMPTY DICTIONARY
count <- frequency.GET(character, 0)
```

or:

```
FUNCTION lookup ACCEPTS(values AS DICTIONARY, key)
RETURN values.GET(key)
ENDFUNCTION
```

or when a class property is declared AS DICTIONARY.

If the receiver is not dictionary-backed, GET remains an ordinary user-defined method name.

Python translation:

```
dictionary.GET(key) -> dictionary.get(key)
dictionary.GET(key, default_value) -> dictionary.get(key, default_value)
```

13.4.2 Backwards Compatibility With v1.0

```
The v1.0 spellings dictionary~GET(key) and dictionary~GET(key, default_value) remain accepte
for backwards compatibility.
Under v1.1 compatibility normalisation they canonicalise to dictionary.GET(key) and
dictionary.GET(key, default_value).
```

The ~GET spelling is accepted but non-canonical in v1.1.

13.5 Dictionary Membership

key IN dictionary

key NOT IN dictionary

14. Random Rules

14.1 Random Integer

```
RANDOM INT(minimum, maximum)
```

Both endpoints are inclusive.

Formal rule:

`random_integer_expression ::=`

“RANDOM INT” ,

“(” ,

`expression ,`

“,” ,

`expression ,`

“)” ;

14.2 Random Float

`RANDOM FLOAT(minimum, maximum)`

Formal rule:

`random_float_expression ::=`

“RANDOM FLOAT” ,

“(” ,

`expression ,`

“,” ,

`expression ,`

“)” ;

14.3 Random Boolean

`RANDOM BOOLEAN()`

Formal rule:

`random_boolean_expression ::=`

“RANDOM BOOLEAN” ,

“(” ,

“)” ;

14.4 Combined Rule

`random_expression ::=`

random_integer_expression

```
| random_float_expression
| random_boolean_expression ;
```

15. Error Handling

15.1 TRY-CATCH

`TRY`

statements

`CATCH error`

statements

`ENDTRY`

The error identifier is optional.

`TRY`

statements

`CATCH`

statements

`ENDTRY`

Formal rule:

`try_statement ::=`

“TRY” ,

newline ,

{ statement | comment } ,

“CATCH” ,

[identifier] ,

newline ,

{ statement | comment } ,

“ENDTRY” ,

`newline ;`

15.2 RAISE

Optional explicit error generation:

`RAISE expression`

Example:

```
RAISE STRING("Invalid input")
```

Formal rule:

```
raise_statement ::=
```

“RAISE” ,

```
expression ,
newline ;
```

16. Class Rules

Class support is included but may be excluded from the first experimental subset if the 50 algorithms do

not require it.

16.1 Class Definition

```
CLASS ClassName
```

declarations

```
ENDCLASS
```

Formal rule:

```
class_definition ::=
```

“CLASS” ,

```
identifier ,
```

newline ,

{ constructor_definition | method_definition | property_declaration | comment } ,

“ENDCLASS” ,

```
newline ;
```

16.2 Constructor

The dialect uses THIS instead of Python’s self.

```
CONSTRUCTOR ACCEPTS(x, y)
PROPERTY THIS.x <- x
PROPERTY THIS.y <- y
ENDCONSTRUCTOR
```

Formal rule:

```
constructor_definition ::=
```

“CONSTRUCTOR” ,

“ACCEPTS” ,

“(” ,

`[ parameter_list ] ,`

“)” ,

newline ,

{ statement | property_assignment | comment } ,

“ENDCONSTRUCTOR” ,

`newline ;`

16.3 Property Declaration

`PROPERTY name`

or:

`PROPERTY name AS INTEGER`

Formal rule:

`property_declaration ::=`

“PROPERTY” ,

`identifier ,`

[“AS” , type_name] ,

`newline ;`

16.4 Property Assignment

`PROPERTY THIS.name <- value`

Formal rule:

`property_assignment ::=`

“PROPERTY” ,

“THIS” ,

“.” ,

`identifier ,`

“<-” ,

`expression ,`
`newline ;`

16.5 Method Definition

```
METHOD name ACCEPTS(parameters)
```

statements

```
ENDMETHOD
```

Formal rule:

```
method_definition ::=
```

“METHOD” ,

```
identifier ,
```

“ACCEPTS” ,

“(” ,

```
[ parameter_list ] ,
```

“)” ,

newline ,

{ statement | comment } ,

“ENDMETHOD” ,

```
newline ;
```

16.6 Property Access

THIS.property_name

```
object.property_name
```

Formal rule:

```
property_access ::=
identifier ,
```

“.” ,

identifier ;

THIS is treated as a reserved contextual identifier inside classes.

16.7 Member-Method Invocation

v1.1 explicitly distinguishes property access from invocation by the presence of parentheses.

Canonical syntax:

```
object.method(arguments)
```

Examples:

stack.pop()

drone.move(nextMove)

`registry.GET(name)`

Formal rule:

`member_call_expression ::=`

primary_expression ,

“.” ,

`identifier ,`

“(” ,

`[ argument_list ] ,`

“)” ;

The method identifier remains case-sensitive.

Parentheses distinguish a method invocation from property access:

```
x.item          COMMENT property access
x.item()        COMMENT zero-argument method invocation
```

A member call may be used as an expression or as a standalone statement.

17. Type Names

`type_name ::=`

“INTEGER”

```
| "FLOAT"
| "STRING"
| "CHAR"
| "BOOLEAN"
| "LIST"
| "DICTIONARY"
| "SET"
| "TUPLE"
| identifier ;
```

A user-defined class name may be used as a type.

18. Statement Grammar

`statement ::=`

assignment_statement

```
| output_statement
| call_statement
| append_statement
| insert_statement
| remove_first_statement
| remove_last_statement
| remove_value_statement
| if_statement
| while_statement
| range_for_statement
| for_each_statement
| return_statement
| break_statement
| continue_statement
| quit_statement
| try_statement
| raise_statement ;
```

19. Declaration Grammar

```
declaration ::=
```

function_definition

```
| class_definition ;
```

20. Literal Grammar

```
literal ::=
```

integer_literal

```
| float_literal
| typed_integer
| typed_float
| string_literal
| char_literal
| boolean_literal
| none_literal
| tuple_literal
| typed_list_literal
| mixed_list_literal
| set_literal
| empty_literal ;
```

21. Complete High-Level Grammar

```
program ::=
{ declaration | statement | comment } ;

declaration ::=

function_definition
| class_definition ;

function_body ::= { function_body_item } ;

function_body_item ::=
statement
| function_definition
| comment ;

statement ::=
assignment_statement
| output_statement
| call_statement
| append_statement
| insert_statement
| remove_first_statement
| remove_last_statement
| remove_value_statement
| if_statement
| while_statement
| range_for_statement
| for_each_statement
| return_statement
| break_statement
| continue_statement
| quit_statement
| try_statement
| raise_statement ;

expression ::=

or_expression ;

or_expression ::=

xor_expression ,

{ "OR" , xor_expression } ;

xor_expression ::=

and_expression ,

{ "XOR" , and_expression } ;
```

```
and_expression ::=
```
not_expression ,
{ “AND” , not_expression } ;
```
not_expression ::=
```
“NOT” , not_expression
```
| comparison_expression ;
comparison_expression ::=
```
additive_expression ,
[
(
“==”
```
| "!="
| "<"
| ">"
| "<="
| ">="
| "IN"
| "NOT IN"
```
),
additive_expression
];
```
additive_expression ::=
```
multiplicative_expression ,
{ (“+” | “-”) , multiplicative_expression } ;
```
multiplicative_expression ::=
```
power_expression ,
{
(
“*”
```
| "/"
| "INTEGER DIVISION"
| "MODULUS"
```

),

power_expression

};

```
power_expression ::=
```

unary_expression ,

[“EXP” , power_expression] ;

```
unary_expression ::=
```

“-” , unary_expression

```
| primary_expression ;
```

```
primary_expression ::=
```

literal

```
| identifier
| indexed_access
| property_access
| member_call_expression
| call_expression
| cast_expression
| isinstance_expression
| input_expression
| length_expression
| join_expression
| dictionary_get_expression
| random_expression
| set_operation
| "(" , expression , ")" ;
```

```
member_call_expression ::=
```

primary_expression ,

“.” ,

```
identifier ,
```

“(” ,

```
[ argument_list ] ,
```

“)” ;

```
cast_expression ::=
"CALL" , cast_type , "(" , expression , ")" ;
```

```
cast_type ::=
"INTEGER"
| "FLOAT"
| "STRING" ;

isinstance_expression ::=
"CALL" , "ISINSTANCE" , "(" , expression , "," , type_reference , ")" ;

type_reference ::=
runtime_type
| "(" , runtime_type , { "," , runtime_type } , ")" ;

runtime_type ::=
"INTEGER"
| "FLOAT"
| "STRING"
| "BOOLEAN"
| "LIST"
| "DICTIONARY"
| "SET"
| "TUPLE"
| identifier ;

quit_statement ::= "QUIT" , newline ;

none_literal ::= "NONE" ;
```

22. Static Semantic Rules

The parser may produce a syntactically valid AST that still violates semantic rules. The following checks

must therefore be performed after parsing.

22.1 Identifier Rules

An identifier must be declared or assigned before it is read.

Invalid:

```
OUTPUT total
total <- 0
```

22.2 Scope Rules

Variables declared within a function belong to that function.

Loop variables belong to the containing function or block, depending on the implementation policy.

Function parameters are available throughout the function body.

22.3 Type Compatibility

Operations must use compatible types.

Valid:

```
total <- INTEGER(5) + INTEGER(2)
```

Invalid:

```
total <- INTEGER(5) + STRING("two")
```

22.4 Boolean Conditions

Conditions used by IF and WHILE must evaluate to Boolean values.

22.5 Return Rules

A RETURN statement may only occur inside a function, method, or constructor where explicitly permitted.

Constructors should not return a value.

22.6 Break and Continue Rules

```
BREAK and CONTINUE may only occur inside loops.
```

22.7 Collection Rules

List indexes must be integers.

Dictionary keys must be hashable in the generated Python representation.

Collection operations must target compatible collection types.

22.8 Function Call Rules

The number of supplied arguments must match the number of required parameters unless optional

parameters are later added.

22.9 Range Rules

The STEP value must not be zero.

Range loops are end-exclusive.

22.10 Character Rules

A CHAR value must contain exactly one character or one valid escape sequence.

22.11 Member Calls and Dictionary GET Resolution

Property access object.property and member invocation object.method(…) are distinct constructs.

A member call is resolved according to the receiver type or other static evidence.

When the receiver is dictionary-backed and the method identifier is GET, the call resolves to the dictionary intrinsic.

The dictionary intrinsic GET accepts exactly one or two arguments.

When the receiver is not dictionary-backed, GET remains an ordinary user-defined method identifier.

An unknown receiver type must not be assumed to be a dictionary merely because the method is named GET.

NONE is a literal value and is exempt from identifier declaration-before-use checks.

22.12 Import Directive Rules

An executable `COMMENT import module` binds the imported module root name from that point onward. When `AS alias` is present, the alias is the bound name.

A `COMMENT from module import name1, name2` directive binds each explicitly listed imported name. A wildcard import is executable but its imported-name set is not statically enumerated by AJ validation.

Import directives do not legitimise unrelated unresolved identifiers and must not be inserted implicitly by the compatibility or compilation stages.

The exact adapter identifier `string_method_punctuation` lowers to `string.punctuation`; the source remains responsible for the explicit `COMMENT import string` declaration where that module binding is required.

22.13 Nested/Local Function Rules

A function declared inside another function belongs to the lexical scope of the enclosing function and must not be promoted to top-level or module scope.

A nested function may call itself recursively by name and may read values already visible in an enclosing lexical scope.

Assignment to a simple identifier inside a nested function follows that nested function local-binding rule. v1.2 introduces no NONLOCAL-style rebinding construct.

Function declarations are not first-class values in v1.2. Callback parameters, function references, anonymous functions, callable- value storage, and closure escape semantics remain outside the dialect.

22.14 Explicit Cast Rules

`CALL INTEGER(expression), CALL FLOAT(expression) and CALL STRING(expression) are explicit ru`

22.15 QUIT Rules

QUIT may occur at top level or within any executable block. It terminates the complete program immediately. QUIT is not equivalent to RETURN, BREAK or CONTINUE and must be represented as a terminal control-flow operation.

22.16 Runtime Type Inspection Rules

`CALL ISINSTANCE(expression, type_reference)` evaluates to BOOLEAN. The value expression is evaluated once. A single type reference tests one runtime target; a parenthesised type list tests whether the value is an instance of any listed target. Built-in type references use Python-compatible `isinstance` semantics in generated Python.

Type references are syntax-only operands and are not ordinary value expressions. No ISINSTANCE test may be inserted implicitly. Within the true branch of a directly evidenced ISINSTANCE condition, the static semantic validator may conservatively narrow an untyped identifier sufficiently to avoid rejecting operations that the condition explicitly proves valid. Multi-type tests do not create a new first-class union type.

23. Normalisation Rules

The evaluation framework may perform conservative normalisation before strict parsing.

Permitted transformations include:

Keyword case normalisation

Unicode arrow ← -> <-

Removal of Markdown code fences

Line-ending normalisation

Trailing whitespace removal

Approved spelling correction

Approved synonym mapping

v1.1 compatibility normalisation may also include:

`Legacy dictionary~GET(...) -> dictionary.GET(...)`

Case normalisation of None / none -> NONE when used as a standalone null literal

These transformations must obey the same logging and semantic-preservation requirements as all other normalisation.

v1.2 compatibility normalisation may also preserve a source-supplied nested function relationship when lexical nesting is structurally recoverable.

It must not hoist a nested helper, manufacture a nested helper, infer callback semantics, or introduce nonlocal-style rebinding that the source did not provide.

Examples:

while x < 5 do

may become:

```
WHILE x < 5 DO
```

x←5

may become:

```
x <- 5
```

The framework must log every transformation.

It must not:

invent missing variables;

insert missing algorithmic operations;

change conditions;

add missing branches;

alter arithmetic operators;

repair semantic logic silently.

v1.3 normalisation must preserve explicit casts and QUIT statements exactly. It must not insert an implicit CALL INTEGER(…), CALL FLOAT(…), CALL STRING(…), or QUIT operation that was absent from the source.

Where a reserved AJ keyword such as INTEGER, FLOAT, STRING, CHAR or QUIT is unambiguously used as an identifier binding in non-canonical source, a source-proven restricted-identifier escape may be applied by the compatibility framework. Such a rename must be logged and must not alter constructor, cast or statement uses of the same keyword.

v1.4 normalisation must preserve explicit `CALL ISINSTANCE(...)` tests and their supplied type references. It must not infer or insert runtime type tests that were absent from the source. `ISINSTANCE` is reserved syntax in v1.4; source-proven identifier escaping remains permitted when binding evidence proves that the same spelling is being used as an identifier.

v1.5 normalisation must preserve exact source-authored COMMENT import directives and must not convert arbitrary comments into imports. It must preserve the exact adapter spelling `string_method_punctuation` when present and must not infer punctuation semantics for other undefined source forms.

24. Guided Link Rules

Guided Links are documentation annotations and do not alter program execution.

Recommended syntax:

```
GUIDED LINK("typed lists")
```

Formal rule:

```
guided_link ::=
```

“GUIDED LINK” ,

“(” ,

quoted_string ,

“)” ,

```
newline ;
```

Guided Links may appear between statements and should be:

- stored as metadata;
- excluded from AST execution nodes;
- excluded from generated Python;
- optionally shown in teaching-oriented visualisations. ## 25. Complete Example

```
FUNCTION find_largest ACCEPTS(numbers AS LIST)
IF LENGTH OF numbers == 0 THEN
```

RAISE STRING(“The list cannot be empty”)

```
ENDIF

largest <- numbers[0]

FOR EACH number IN numbers DO
IF number > largest THEN
largest <- number
ENDIF
ENDFOR

OUTPUT STRING("The largest value is {largest}")
RETURN largest
```

ENDFUNCTION

26. Expected AST Outline

ProgramNode

FunctionNode: find_largest --

ParameterNode: numbers AS LIST |–

Body --

IfNode |–

Condition | |–

EqualNode | |--

LengthNode | | |–

IdentifierNode: numbers | | | --

IntegerLiteralNode: 0 | | --

ThenBody | --

```
RaiseNode
|
\--
StringLiteralNode
|
\--
AssignmentNode
|--
IdentifierNode: largest
|
|--
IndexAccessNode
|
\--
IdentifierNode: numbers
|
|--
IntegerLiteralNode: 0
|
\--
ForEachNode
|--
```

```
Iterator: number
||--

Collection: numbers
||--

Body
|\--

IfNode
|
\--

GreaterThanNode
|
|--

AssignmentNode
|
\--

OutputNode
|--

ReturnNode
\--
```

27. Versioning Rule

This specification is frozen as:

Cambridge-Style Pseudocode - AJ_24038155 Dialect v1.5

v1.5 supersedes the v1.4 specification by integrating the approved import-directive and standard-library-adapter language changes while retaining all v1.0, v1.1, v1.2, v1.3 and v1.4 rules:

NONE as a native null literal;

general member-method invocation using object.method(…);

canonical dictionary retrieval using dictionary.GET(…);

backwards-compatible acceptance of the v1.0 dictionary~GET(…) form;

nested/local FUNCTION declarations inside function bodies with lexical scope preservation;

explicit runtime casts using CALL INTEGER(expression), CALL FLOAT(expression) and CALL STRING(expression);

whole-program termination using QUIT;

explicit runtime type inspection using CALL ISINSTANCE(expression, type_reference), including multi-type tests such as (INTEGER, FLOAT);

exact executable import directives using COMMENT import … and COMMENT from … import …;

the exact standard-library adapter string_method_punctuation -> string.punctuation;

callbacks and first-class function values remain outside v1.5.

Typed literal constructors retain their previous meanings. Type references used by ISINSTANCE are syntax-only and do not become ordinary first-class values. Ordinary COMMENT text remains non-executable unless its complete payload matches a v1.5 import directive. Implicit casts, implicit runtime type tests and implicit imports are not introduced.

Any later syntax addition, removal, or semantic alteration must create a new version, for example:

AJ_24038155 Dialect v1.6

AJ_24038155 Dialect v2.0

AI outputs should always be evaluated against the exact grammar version recorded at generation and evaluation time.